Auditing a Released Back-Translation Evaluator for Sign Language Production: Sensitivity to Training-Pool Replay and Checkpoint Reconstruction

[1*]

[1*].

Abstract

Back-translation (BT) evaluators mediate sign language production rankings, but a released checkpoint may yield orderings that do not survive retraining or reference replacement. On the 641-sequence PHX-public subset of SLRTP2025, a constructed whole-donor replay probe (TN-PURE-v1) scores 23.02 sacreBLEU under the released BT evaluator versus 12.78 for the recorded test poses, a gap of **+10.24** BLEU points (95% CI $[\mathbf{+8.89, +11.58}]$). Across 43 independently trained checkpoints (dev BLEU-4 4.5–10.9, all less competent than the released evaluator's 13.38), none reproduces a comparable gap: all 43 have non-positive PURE–GT gaps (range $[\mathbf{-2.01, 0.00}]$), decoded with a single canonical donor registry. On the matched 461-item confidence=1 subset of Czehmann et al.'s human back-translations, reference replacement attenuates the gap from **+10.03** to **+0.95** (90.5%). The released evaluator also decodes its 7,060-item training pool at 78.8 BLEU (70.7% exact match) while retaining non-trivial dev/test performance (13.38/12.78); pose–text permutation, input-side pose shuffle with output-equivariance verification, stratified exact-match, and same-signer held-out controls rule out a pose-ignorant pure ID/cache lookup. The released evaluator's training log shows validation PPL rising from 14.84 at step 546 to 25.42 at the best-BLEU step 1820 (final step 2828: 30.51), indicating that decoded-BLEU selection chose a checkpoint in a higher-NLL weight-space region that NLL-selected reconstructions never enter. This audit is scoped to the public training recipe: competence-matched replication was not possible, and the specific mechanism is not identifiable from public artifacts.

Keywords: sign language production, back-translation evaluation, retrieval-augmented generation, multimedia quality assessment, evaluator non-replication

1 Introduction

Sign language production (SLP) systems are commonly evaluated by back-translation (BT): a trained sign recognizer decodes each output pose sequence, and a text metric compares the decoding with the reference. This measures recognizer-readable content through a learned measurement model, so system rankings may reflect both the evaluated outputs and the inductive biases of a particular evaluator checkpoint. When proposing BT evaluation for SLP, Saunders et al. Saunders et al. (2021) argued that changing the BT model changes absolute scores but preserves relative rankings and recommended BT for comparing between similar model configurations. The present audit tests a stress case of that claim, but because all constructible checkpoints in our family are less competent than the released evaluator (dev BLEU-4 4.5–10.9 vs. 13.38), our non-reproduction result is scoped to the public training recipe and does not constitute a strong test of their rank-stability proposition.

Prior work shows that BT scores can be modest for recorded real signing and can improve without faithful target conditioning Walsh et al. (2024); Hong and Košecká, Jana (2026), and sign-native or pose-aware metrics provide complementary evidence Kim et al. (2024); Imai (2025); Jiang et al. (2026). Nevertheless, BT remains central to benchmarks such as SLRTP2025 Walsh et al. (2025). An unresolved question is whether a retrieval–reference stress-test ordering persists across independently trained instances of the same BT architecture.

This question is exposed by a retrieval–reference reversal on PHX-public: under the original SLRTP2025 evaluator, source-conditioned whole-donor retrieval scores 23.02 sacreBLEU (0–100 scale), whereas the recorded test poses score 12.78. This neither establishes better signing by retrieval nor invalidates BT evaluation in general — it asks whether the reversal persists across evaluator checkpoints or depends on properties of the original evaluator.

We address this question in four stages. First, we compare the reversal with reference-pose, generative, and random-donor controls under the original evaluator, verifying protocol details (scorer equivalence, frame-rate handling, decoding parameters). Second, we evaluate the same fixed outputs with fourteen independently trained recipe-conditioned reconstructions and quantify paired uncertainty, documenting that no rescue variant, checkpoint-selection objective (NLL, decoded BLEU, decoded WER), or step-faithful run reaches competence parity. Third, we test reference validity: on the matched 461-item confidence=1 subset of public human sign-to-text back-translations Czehmann et al. (2026), the reversal attenuates from +10.03 to +0.95 BLEU points (90.5%). Fourth, we characterize what distinguishes the released pipeline via five diagnostic views of a training-content readout property. Evaluator-aware oracle analyses are in SI Sup. H because their unmasked advantage does not survive reference-overlap exclusion (Fig. 1).

Why a constructed probe matters.

Retrieval-augmented SLP entries draw donors from the evaluator's training pool, so the TN-PURE-v1 stress test isolates a risk that production systems face in diluted

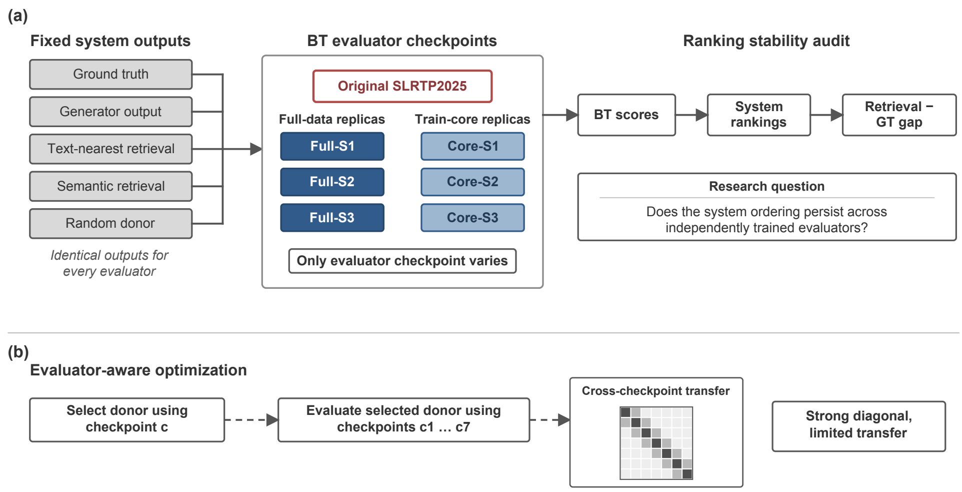

Fig. 1: Overview of the multi-checkpoint evaluation audit. (a) A fixed set of reference-pose, generative, retrieval, and random-donor outputs is evaluated by the original SLRTP2025 BT checkpoint and fourteen independently trained recipe-conditioned reconstructions. (b) Evaluator-aware analysis: donors selected by maximizing likelihood under one checkpoint are evaluated by all seven checkpoints.

form. Methodologically, the audit contributes a template — provenance table, checkpoint registry, evaluator-replication controls — that other benchmarks can adopt for rank-stability testing.

2 Related Work

Learned evaluation in sign language production.

Neural SLP maps spoken language or glosses to pose and often evaluates with a frozen sign recognizer Camgoz et al. (2018); Saunders et al. (2020). BT was proposed as an SLP evaluation protocol by Saunders et al. Saunders et al. (2021), who argued the choice of BT model shifts absolute scores but not relative rankings; our audit is a direct stress test of that rank-stability claim. SignBLEU Kim et al. (2024), SiLVERScore Imai (2025), and pose-native measures offer complementary views, but BT BLEU remains central to benchmarks such as SLRTP2025 Walsh et al. (2025).

Retrieval and shortcut risks.

Retrieval-based SLP ranges from phonetic composition to retrieval-enhanced generation Walsh et al. (2024,?); Zuo et al. (2025); Tasyurek et al. (2025). The SLRTP2025 report describes a leading retrieval-based entry, making donor copying pertinent Walsh et al. (2025). More generally, memorization in retrieval-augmented language modeling shows that strong task scores can reflect reproduction of training examples Carlini et al. (2021); Kandpal et al. (2022); Khandelwal et al. (2020). A shortcut observed under one learned evaluator does not show that the resulting ordering persists under another checkpoint.

Robustness of learned metrics and corpus audits.

Hong and Košecká Hong and Košecká, Jana (2026) show BT-BLEU can diverge from target faithfulness; Walsh et al. Walsh et al. (2024) document a low ground-truth BT baseline; Jiang et al. Jiang et al. (2026) recommend BT likelihood over decoded BLEU. Yazdani et al. Yazdani et al. (2026), BackTranslation2.0 Cory et al. (2026), and He et al. He et al. (2024) expose further fragilities of model-based generation metrics. Two 2026 PHOENIX-2014T audits bear directly on our reversal: Czehmann et al. Czehmann et al. (2026) released human back-translations and documented reference information loss; Alkain et al. Alkain et al. (2026) showed PHOENIX-2014T has the highest train–test text overlap among tested SLT corpora. Artiaga et al. Artiaga et al. (2025) showed signer-dependent evaluation overestimates SLT capability. None of these studies runs multi-checkpoint evaluator experiments or donor-pool substitution controls.

3 Methods

3.1 Evaluation design

Let E_c denote a BT evaluator checkpoint and $o_{i,n}$ the pose sequence for output condition i and item n. Checkpoint c produces $h_{c,i,n} = E_c(o_{i,n})$ and $B_{c,i} = M(\{(h_{c,i,n}, y_n)\}_{n=1}^{N})$, where y_n is the reference text and M is the corpus-level sacreBLEU functional unless stated otherwise. The values $B_{c,i}$ induce a ranking over fixed output conditions. We call a result *evaluator-pipeline-sensitive* when a substantive pairwise ordering changes across independently trained checkpoints under a fixed output set. The fixed primary set comprises recorded reference poses (GT), random donor copy, source-conditioned whole-donor retrieval, a Progressive Transformer generator, and retrieval–generator compositions; oracle-gloss variants are diagnostic controls.

The main evaluation uses PHX-public, a German weather-domain text-to-pose benchmark. The released SLRTP2025 pose records contain 7,060/515/641 sequences; one official test ID is absent from the released materialization, so every result is conditioned on the complete 641-record release (SI Sup. A). Table 1 gives the three-way provenance: the official PHOENIX-2014T split, the released SLRTP2025 pose materialization, and the released evaluator's actual training manifest (confirmed via `config.yaml` which references the same `train.pickle/dev.pickle/test.pickle` that ship as `train.pt/dev.pt/test.pt`). Supplementary analyses use **PHX-holdout-MSKA** (133-joint, 1,538 windows) and CSL-Daily (Mode B, 1,176 sequences; SI Sup. M). Table 3 summarizes the protocols.

All pose sequences are evaluated at 12.5 fps, matching the SLRTP2025 evaluator training distribution (`skeleton_subsample=2`). Every canonical pose bundle is *stored* at 12.5 fps after exactly one construction-time `[::2]` transform; the audit bypasses the released wrapper and calls the BT model directly, so no double subsampling occurs. A three-path regression confirms this: canonical and explicit 25→12.5-fps paths give

Table 1: Three-way data provenance for PHX-public. The released evaluator's training manifest matches the released pose materialization (7,060/515/641), not the full official PHOENIX-2014T split. All 41 missing IDs share the reason `absent_from_slrtp2025_released_pose_records` (SI Sup. A). SHA-256 hashes verify file integrity.

Split source	Train	Dev	Test
Official PHOENIX-2014T	7,096	519	642
Released SLRTP2025 pose (`.pt`)	7,060	515	641
Released evaluator `config.yaml` manifest	7,060	515	641
Missing from official	36	4	1

Table 2: Experimental protocol summary.

Setting	Recognizer	Layout	Split size	N eval	α
PHX-public	SLRTP2025 evaluator	178-joint	641 seq	641	0.88 (tuned)
PHX-holdout-MSKA	mska_slr (phoenix-2014t)	133-joint	1,538 win	1,538	0.50 (frozen)
CSL-Daily	mska (csl-daily)	133-joint	256 win	256	0.50 (frozen)

identical GT BLEU-4 (12.78) and WER (85.77), while an erroneous second subsample degrades them (6.65 / 86.85; full table in SI). The released BT model applies *no* internal subsampling: its `subsample` attribute is set but never used. The official Codabench starter-bundle PT scores 1.59 BLEU under the released evaluator, matching the official 1.5942. PHX-public uses α=0.88 (selected on dev); PHX-holdout-MSKA and CSL-Daily use α=0.5.

3.2 Systems and retrieval controls

For each target, source-conditioned whole-donor retrieval scans the complete training pool after Unicode NFKC normalization, whitespace normalization and lowercasing, and selects the sequence with maximum source-text token-set Jaccard similarity. Oracle-gloss retrieval instead uses target gloss and is treated as an input-leakage diagnostic; random retrieval selects a donor uniformly. Donor exclusion removes exact normalized text or gloss duplicates or donors above threshold τ. Gloss-based exclusion uses target gloss and is therefore an offline diagnostic rather than a deployable inference procedure.

Text-nearest tie-breaking.

Jaccard ties are resolved by minimum character-level Levenshtein distance and then by minimum SHA-256 hash of the training key. The §3.2 algorithm also excludes donors whose NFKC-normalized text equals the query's (exact-normalized-text exclusion).

Two materializations. The headline observations throughout this paper (PURE 23.79, gap +11.01; Tables 5, 4, 6, 8) use an earlier frozen materialization that predates the exclusion fix. The 29-checkpoint re-decode panel and the distillation ladder (Table 13) use the full §3.2 algorithm with exclusion, yielding PURE 23.02 (gap +10.24). The donor-registry consistency note (§4.4) quantifies the impact: 610/641 items produce identical output; on the 31 differing items, a post-hoc candidate search found that 30 had at least one excluded Jaccard=1 candidate whose decode matched the earlier hypothesis, consistent with the exclusion rule explaining most differences. Both materializations are deterministic; the §3.2 algorithm is reproduced in the artifact at `scripts/build_canonical_panel.py`.

Retrieval composition.

TN-PURE-v1 uses whole-donor copy with [::2] subsampling to 12.5 fps, preserving the donor's original duration. TN-PTCOMP-v1 (a secondary control) composes a PT scaffold with donor hands and face via α-blending of upper-body joints (α=0.88 for PHX-public; full joint-set definitions and duration/word-count matching statistics in the artifact). The separately matched seen–unseen factorial retains 605/641 targets matched on duration and word count.

Donor-origin contrast retrieval systems.

To test whether the retrieval advantage co-occurs with donor origin, we construct frozen probe systems from the same 641 queries: **UNSEEN-PURE-v1** retrieves from the 640 other test poses (excluding the query itself, its sequence root, and exact-text matches); **CROSSFIT-PURE-A/B-v1** split queries by hash into 331/310 folds, each retrieving from the other fold; **SEEN-PURE-MATCHED-v1** selects a train-pool donor within ±0.1 Jaccard of each UNSEEN donor (622/641 admit one); **SEEN-RAND640-MATCHED-v1** applies matched selection inside frozen 640-donor subpools. SEEN-PURE-v1 re-uses the canonical TN-PURE-v1 registry. Full construction rules (calipers, tie-breaks, signer constraints) are in the artifact; these are constructed probes, not deployable generators.

3.3 Evaluator checkpoint family

The primary inferential checkpoint family comprises the original frozen SLRTP2025 evaluator and six *recipe-conditioned reconstruction* runs trained from scratch with the documented architecture and a common frozen train/development partition (primary seeds 101–606; eight extension seeds 707–1405 use the identical protocol). The original bundle's `validations.txt` logs 202 validation points over 2,828 optimizer steps (≈102 epochs), with decoded-BLEU selection reaching best dev BLEU-4 13.38 at step 1,820; intermediate checkpoints are not published. We use two reconstruction protocols: *paper-derived* (14 runs, legacy script, 300 epochs, NLL selection) and *released-config* (4 runs, up to 3,000 epochs, decoded-BLEU early stopping). Both share the documented architecture (3L/256/512ff, 8 heads), tokenizer, manifests (7,060/515), [:: 2] subsampling, Adam (1e-3, betas .9/.998, wd .001), batch size 256, and plateau ×0.8 LR schedule (full provenance in SI). A total of 70 training runs were completed across 13 families; 67 have dev metrics, 52 non-distillation runs enter the competence gate, and

43 have canonical PURE–GT gap under the §3.2 donor registry. Five checkpoints produce degenerate output (three α=1.0 students: 641/641 or 483/641 empty hypotheses; two smallest ladder fractions: BLEU=0.00) and are excluded from the dose–response range display.

Readout-with-generalization signature (post-hoc operational definition).

A checkpoint exhibits the readout-with-generalization signature if it has both (i) training-pool free-decode BLEU $\geq$ 50 *or* exact-match $\geq$ 50%, and (ii) dev BLEU-4 $\geq$ 8.0. These thresholds were defined after observing the released evaluator (78.8 BLEU / 70.7% EM, dev 13.38) vs. the reconstruction family (max 11.5 train / 9.9 dev, no simultaneous satisfaction). The thresholds are operational heuristics for this case study, not validated criteria.

3.4 Metrics and uncertainty

The recorded test poses (GT) are decoded by each BT recognizer and compared with the original reference text. We report corpus BLEU-4 on the 0–100 sacreBLEU scale as the primary metric Post (2018); sentence-level statistics (pass-through, slot-F1) are reported separately on the 0–1 scale. BLEU-1, chrF, ROUGE-L, BERTScore Zhang et al. (2020) (`bert-base-multilingual-cased`, no baseline rescaling, F1), BLEURT Sellam et al. (2020) (`Elron/bleurt-large-512`, a PyTorch-native substitute for BLEURT-20 with $r = 0.87$ rank correlation and -0.47 bias against canonical BLEURT-20 on 100 German pairs), and WER (jiwer 3.1.0 normalized) provide complementary decoded-text controls; BT likelihood and pose-DTWp provide non-decoded controls.

Scorer equivalence.

The vendored official scorer (SacreBLEU 1.4.2) and the paper scorer (SacreBLEU 2.5.1, `tok:13a`, `smooth:exp`) give identical corpus BLEU on all 28 evaluation cells (maximum $|\Delta| = 0.0000$; byte-identical hypotheses, all n-gram orders non-zero). We report the 2.5.1/13a signature as primary.

We report two forms of uncertainty. *Item-sampling uncertainty* uses paired sequence-level bootstrap (10,000 resamples, seed 42, percentile 95% CIs): per item, n-gram sufficient statistics are precomputed once, and every resample re-aggregates them into a corpus BLEU. A pairwise difference is called detectable when its interval excludes zero. Significance testing of the primary estimand uses a paired approximate-randomization permutation; because the headline gap was first observed in exploratory analysis and the estimand was frozen afterwards, we report a descriptive Monte-Carlo estimate: 0 of 10,000 permutations exceed the observed gap ($p \leq 10^{-4}$, exact 95% binomial interval $[0, 0.00037]$). *Training-replicate uncertainty* is reported only for the recipe-conditioned population: a descriptive seed-level 95% Student-t prediction interval (not extrapolable to the released evaluator). The eight extension seeds are post-hoc additions and enter only that interval, never the six primary comparisons. The single primary estimand is the PURE–GT decoded-sacreBLEU gap under the released evaluator; all other contrasts are secondary and are Holm-corrected within a defined family or explicitly labelled descriptive (SI Sup. C).

Table 3: Decoded-text metric sensitivity under the original SLRTP2025 evaluator (641 sequences; beam 3, length penalty -1, max length 30) under the official implementations (§4.1). The reversal direction is consistent across decoded-text metrics; teacher-forced NLL and pose-DTWp do not reproduce it.

System	BLEU-4	BLEU-1	chrF	ROUGE-L	WER	BERTSc.
GT-v1 (reference poses)	12.78	34.40	34.59	35.20	85.77	0.757
PT-v1	1.59	16.91	19.03	17.26	101.68	0.669
TN-PTCOMP	18.86	46.35	41.88	45.17	77.36	0.793
TN-PURE	**23.02**	**55.43**	**47.82**	**53.07**	**71.87**	**0.822**
Gap PURE−GT	+10.24	+21.03	+13.23	+17.87	−13.90	+0.066

4 Results

4.1 Retrieval reverses the reference ranking under the original evaluator

Under the original SLRTP2025 evaluator, source-conditioned whole-donor retrieval scored 23.02 sacreBLEU (0–100 scale), compared with 12.78 for the recorded test poses — a retrieval–ground-truth gap of +10.24 BLEU points (95% CI $[+8.89, +11.58]$; 10,000 sequence-level resamples). This fixed-checkpoint reversal motivates the analysis but does not indicate superior sign production.

Controls localized the observation to source-conditioned donor retrieval (Table 5): pure donor copy exceeded the oracle-gloss composed variant (11.8), canonical PT-composed (18.86), PT baseline (1.59), and 20-seed random-donor baselines (0.90 PURE, 0.77 COMP). Oracle-gloss retrieval (which uses reference gloss and is leakage) did not exceed the recorded test poses (11.4–11.8 vs 12.78). Token-normalized teacher-forced NLL (lower is better) does not reproduce the reversal: GT 3.060 vs PURE 3.479, COMP 3.411, PT 4.074. Likewise the normalized pose-DTW diagnostic does not validate semantics (GT 0 by construction vs PT/PURE/random 0.00678–0.00830). The reversal is visible under complementary text metrics on the same hypotheses (Table 4): sacreBLEU-4 +10.24, BLEU-1 +20.94, chrF +12.93, ROUGE-L +17.87, WER −13.90 (better), BERTScore-F1 +0.066. We describe the observation as a *decoded-text-metric reversal* and use sacreBLEU-4 as the primary decoded metric.

Metric implementations.

All secondary metrics use the official SLRTP2025 implementations: they reproduce the official repository values (BLEU-4 12.777, chrF 34.585, WER 85.775, ROUGE 35.196 Walsh et al. (2025)) exactly. Implementation choices matter for chrF (corpus-level sacrebleu `CHRF` defaults; a sentence-level variant yields 48.05 instead of 34.59) and WER (jiwer 3.1.0 with official normalization; raw whitespace-split yields 79.26 instead of 85.78). BLEU-4 is invariant; we fix the official implementations throughout (ablations in SI).

Table 4: PHX-public conditioning and construction controls (641 sequences; 12.5 fps). Columns distinguish whole-donor copy from PT-scaffold composition (α=0.88); all rows share the query set, donor pool, evaluator, and sacreBLEU protocol. Semantic retrievers (LaBSE 19.2/8.4, multi-SBERT 16.5/8.1) lie between the canonical retriever and oracle gloss; full rows in the artifact.

Conditioning	Pure donor copy	PT+retrieval composed
Source text (German Jaccard; original frozen materialization)	**23.02**	18.86
Oracle gloss (input leakage)	11.4	11.8
Random donor (seeds 0–19)	0.90 (SD 0.24)	0.77 (SD 0.19)

Decoding-parameter sensitivity.

Re-decoding across beam size $\{1, 3, 5\}$, length penalty $\{-1, 0\}$, and max length $\{30, 50\}$ shows the reversal is not an artifact of decoding: the PURE–GT gap is +9.9 (greedy), +10.2 (beam 3), +10.8 (beam 5), with length penalty and max length negligible.

Pure donor copy outperformed the composed variant for every source-conditioned retriever, and source-text Jaccard produced the highest pure-donor score (23.02, ahead of LaBSE 19.2 and multi-SBERT 16.5), while oracle-gloss retrieval did not exceed the recorded test poses in either construction (11.4–11.8 vs 12.78): the reversal does not require oracle-gloss input leakage, and pure copy — not composition — is its strongest form.

4.2 Independent evaluator reconstructions do not reproduce the reversal magnitude

TN-PURE-v1 yields an original-evaluator gap of +10.24 BLEU points (95% CI $[+8.89, +11.58]$). Across fourteen reconstruction runs, all fourteen have negative PURE–GT gaps (range $[-1.85, -0.80]$). TN-PTCOMP-v1 yields an original gap of +6.09 $[+4.76, +7.41]$; all fourteen reconstruction estimates are negative. The seed-level 95% prediction interval is $[-1.94, -0.46]$ (mean -0.55, SD 0.39, 2/14 positive). For the six primary runs, original-minus-run difference-in-differences estimates range from +10.89 to +11.99; every 95% interval excludes zero.

4.2.1 Recipe-conditioned evaluator reconstructions

All fourteen reconstructions share frozen manifests, architecture, hyperparameters, and selection rule; seeds 101–1405 vary initialization and data order (Table 6). All evaluators are decoded with the canonical §3.2 donor registry (exact-normalized-text exclusion applied).

Competence is assessed on the released 515-sequence development subset, never on PHX-public for checkpoint selection. Original obtains dev BLEU-4 13.38 and WER 83.37 (official jiwer implementation, 0–100 scale); reconstruction runs obtain 6.69–9.07 and WER 88.21–94.83. Their selected validation NLL is 2.641–2.712 versus Original teacher-forced dev NLL 3.235, showing that likelihood and decoded recognition

Table 5: Canonical fixed-output evaluation on the released 641-pose PHX-public subset, decoded with the §3.2 donor registry. PURE is TN-PURE-v1; COMP is TN-PTCOMP-v1 (frozen materialization). Gaps are relative to local GT (BLEU points). Per-cell values in the artifact.

Evaluator	GT	PURE	Gap	COMP	Gap
Original	12.78	23.02	**+10.24**	18.86	+6.09
Seed 101	9.24	8.45	−0.80	7.92	−0.65
Seed 202	8.22	6.92	−1.30	7.54	−1.28
Seed 303	9.10	7.25	−1.85	8.71	−0.96
Seed 404	9.52	7.71	−1.81	6.38	−0.81
Seed 505	9.71	8.44	−1.28	7.97	−1.08
Seed 606	7.55	6.31	−1.24	7.93	−0.16
Extension 707–1405 (range)	8.05–9.87	6.86–8.97	−1.48 to −0.87	6.90–8.83	−1.03 to −0.44

competence are not interchangeable. On PHX-public, reconstruction GT scores (7.19–9.67 BLEU points) likewise remain below Original 12.78. Consequently, evaluator competence and unavailable original-training details remain confounded with evaluator identity; the experiment identifies evaluator-pipeline sensitivity, not a causal checkpoint-identity effect.

Competence gate and trajectory analysis.

A checkpoint is competence-matched only if both $|\Delta\text{dev BLEU-4}| \leq 1.0$ and $|\Delta\text{dev WER}| \leq 3.0$ (the released evaluator's dev values are 13.38 / 83.37). These are prespecified operational thresholds; we do not claim they are domain-standard. None of the 52 non-distillation trained checkpoints passes either half: every checkpoint fails both (dev BLEU-4 falls short by 3.4–8.9; dev WER exceeds the 86.37 bound by 0.08–13.3). Along dense-checkpoint trajectories (170 saved epochs across eight seeds), no saved epoch reaches the ± 1.0 BLEU band (max 9.73) and the best WER (86.41) still exceeds the 86.37 bound.

The multi-objective Pareto frontier over every saved epoch under three selection objectives (dev NLL, decoded BLEU, decoded WER) also fails to reach the gate. We expanded the rescue search beyond learning rate: batch size $\{128, 512\}$, dropout $\{0.2\}$, weight decay $\{0\}$, label smoothing $\{0.1\}$, and a longer-budget variant (600 epochs, lr 5×10^{-4}), two seeds each (Table 7). All twelve new runs fail the gate (dev BLEU-4 6.8–9.9); the best (weight-decay 0, seed 202, dev BLEU-4 9.92) yields PHX-public GT 9.88 / PURE 8.29 (gap −1.59), extending the dose–response toward the gate without any reversal reappearing. The gate conclusion is insensitive to threshold: across a $\pm[0.5, 2.0]$ BLEU $\times$ $\pm[1.5, 4.5]$ WER grid (12 combinations), 0 of 52 trained checkpoints passes any. We frame the result as released-evaluator pipeline non-reproduction rather than checkpoint identity.

What the released training log establishes.

The bundle's `validations.txt` logs dev BLEU-4 reaching 10.01 at step 770, 12.2 at step 1,134, and 13.38 at step 1,820 (Fig. 2). All constructible reconstructions reach

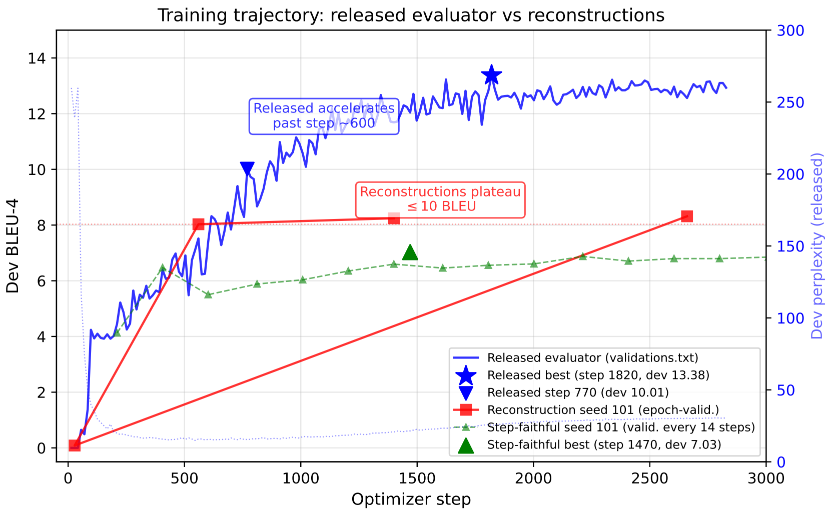

Fig. 2: Training trajectories: released evaluator (blue, decoded-BLEU selection) vs reconstruction seed 101 under epoch-validation (red) and step-faithful protocol (green). The released `best.ckpt` corresponds to step 1,820 (best dev BLEU-4 13.38, blue star). All six primary reconstruction seeds plateau below 8.5 at selection; the released evaluator's dev perplexity bottoms around step 750 and rises thereafter, confirming decoded-BLEU rather than likelihood selection.

8–10 by epoch 50–100 and plateau, while the released evaluator continues improving past epoch 30–60. The training dynamic that produced this slope is not explained by public artifacts.

Competence matching by input degradation.

Spatial jitter ($\sigma \leq 8\%$) barely moves either score; temporal downsampling steadily compresses both. At $k=8$ (one frame in eight), GT reaches 8.75 (inside the reconstruction range), yet PURE still scores 11.69 (gap +2.94). The replay advantage attenuates with input quality but is not eliminated, bounding the competence component without identifying it, since degradation also constitutes a distributional shift. The full degradation table (8 rows) is in SI Sup. F.

The six primary seeds carry precomputed paired-bootstrap intervals; the eight extension seeds reproduce the same pattern (seed-level 95% PI [−1.94, −0.46]).

Cross-metric consistency across checkpoints.

Table 8 reports the gap on the same fixed output under each of the seven primary evaluators across metrics. Under the original, the gap is positive and large for all decoded-text metrics; teacher-forced NLL is more negative for PURE than GT (−0.419 sign-flipped), consistent with the likelihood-vs-decoded-metric divergence Jiang et al. (2026). Under the six reconstructions the reversal disappears: all chrF and ROUGE-L gaps are negative, all WER gaps favor GT, and sacreBLEU-4 gaps are all negative

Table 6: Competence rescue on the released 515-sequence dev split: 20 runs across six hyperparameter families (two to four seeds each; per-seed values in the artifact). All fail the competence gate ($|\Delta$dev BLEU-4$| \leq 1.0$ and $|\Delta$dev WER$| \leq 3.0$; the released evaluator is 13.38 / 83.37; official WER ranges: lr-rescue 87.5–93.1, expanded rescue 86.6–93.1); a long-schedule run reaches a family-maximum 10.02 and still fails. The competence gate conclusion is insensitive to threshold choice: across a $\pm[0.5, 2.0]$ BLEU $\times$ $\pm[1.5, 4.5]$ WER grid of twelve combinations, 0 of 52 trained checkpoints passes any of them.

Variant family	dev BLEU-4 range	min $\|\Delta$BLEU-4$\|$ vs Original
lr 5×10^{-4} / 2×10^{-3} (8 runs)	7.9–9.7	3.7
batch size 128 / 512	6.8–9.2	4.2
dropout 0.2	6.9–7.5	5.9
weight decay 0	**9.8–9.9**	**3.5**
label smoothing 0.1	6.8–7.4	6.0
600 epochs, lr 5×10^{-4}	8.2–8.3	5.1

(range $[-1.85, -0.80]$). The reversal is a property of decoded-text metrics under the original evaluator, not specific to sacreBLEU-4.

Rank stability (see SI Sup. H).

On a fixed nine-system stress-test set, the released evaluator ranks TN-PURE first; five of six reconstructions rank GT first, with moderate cross-checkpoint agreement (mean τ_b=0.44, ρ=0.57).

Reference-validity sensitivity: matched-subset analysis.

Two 2026 corpus audits bear directly on this reversal. Czehmann et al. Czehmann et al. (2026) released manual sign-to-text back-translations of the test set (one deaf fluent L2 DGS signer; per-item confidence flags), documenting information loss, lexical errors, and translationese in the original speech-transcribed references, and showed that reference replacement already changes BLEU scores and rankings on PHOENIX-2014T. Alkain et al. Alkain et al. (2026) quantified train–test text overlap across sign-language translation corpora, showing that a small fraction of highly training-like test items disproportionately inflates corpus BLEU. We reproduce the overlap pattern locally: the 32 test items whose reference exactly matches a training text score GT 40.26 / PURE 78.73 (gap +38.5), while the remaining 609 items score GT 11.93 / PURE 22.20 (gap +10.3).

Because Czehmann et al. compared original-reference and human-reference conditions on the same confidence=1 subset (462 of 642 items, excluding low-confidence, partial, or untranslatable items), we follow the same protocol. On the matched 461-item subset (after join with our 641 released test IDs), the original-reference gap is +10.03 BLEU points (GT 14.94 / PURE 24.98), and the human-reference gap is +0.95 (GT 7.35 / PURE 8.30), a paired attenuation of 9.64 BLEU points (90.5%,

Table 7: PURE–GT gap across metrics and evaluators on PHX-public (641 sequences; canonical decoding) for the original evaluator and the *six primary* reconstructions (of fourteen; the eight extension seeds show the same near-zero/negative pattern in Table 6). chrF, WER and ROUGE-L use the official SLRTP2025 implementations (§4.1). WER and NLL are sign-flipped so positive always means PURE is better than GT. BERTScore-F1 uses `bert-base-multilingual-cased` (bert-score 0.3.12, `lang=de`, `idf=False`, no baseline rescaling); a column is in the artifact. The large reversal is consistent in direction across decoded-text metrics under the original evaluator and disappears under all six primary reconstructions; residual reconstruction signs are mixed and near zero for n-gram metrics.

Evaluator	ΔBLEU-4	ΔBLEU-1	ΔchrF	ΔWER†	ΔNLL†	ΔBERTSc.
Original	+10.24	+21.03	+13.23	+13.90	−0.419	+0.066
Seed 101	−0.80	−1.23	−1.44	−2.95	−0.219	−0.005
Seed 202	−0.98	−0.21	−1.94	−3.52	−0.233	−0.006
Seed 303	−0.71	−0.10	−1.20	−1.53	−0.240	+0.001
Seed 404	−0.36	−0.14	−1.20	−1.87	−0.194	−0.001
Seed 505	−0.89	−0.17	−1.78	−3.41	−0.268	−0.004
Seed 606	−1.24	−1.26	−1.55	−3.46	−0.169	−0.001

†WER and NLL are sign-flipped: positive means PURE is better than GT (lower WER, lower NLL).

bootstrap CI $[7.44, 10.86]$; Table 9). The full-641 comparison (+10.24 → +0.86, attenuation 91.6%) is reported as sensitivity analysis; it mixes reference replacement with floor effects from low-confidence items and is not the primary estimand. This is a *descriptive contrast of two corpus gaps*, not a causal attribution. Under all six reconstruction checkpoints the gap against human references is negative (−0.15 to −0.65). The asymmetry is consistent across metrics on the matched subset: chrF attenuation 82.2% (+12.73 → +2.27), BLEU-1 attenuation 84.0% (+20.01 → +3.19). Czehmann et al. also showed BLEURT is less sensitive to reference replacement than BLEU; our BLEURT substitute (artifact) on the full-641 set gives a gap of +0.165 under original refs and a smaller value under human refs (artifact), consistent with their finding.

A natural concern is that the attenuation is a symmetric floor effect. On the matched subset, switching from original to human references, GT drops 7.59 BLEU-4 points (50.8%) but PURE drops 17.24 points (66.6%) — PURE loses a larger fraction (ratio 1.31×). The asymmetry is even larger under BLEU-1 and chrF, so simple symmetric compression is insufficient, but the asymmetry does not discriminate between reference-text defects and training-distribution alignment.

4.3 Donor-origin contrast characterizes the reversal's co-occurrence with training-pool donors

If the released-evaluator reversal requires exposure to training-pool donors, then restricting retrieval to poses the released evaluator has not seen should remove or

Table 8: Matched-subset reference sensitivity (corpus sacreBLEU, BLEU points) under the released evaluator. Primary analysis: confidence=1 subset (461 items) with original and human references computed on the *same* items. Sensitivity analysis: full 641 items. Human references: Czehmann et al. Czehmann et al. (2026) back-translations (CC BY-NC-SA 4.0). Confidence breakdown: 461 confidence=1.0, 121 confidence=0.5, 59 confidence=0.0.

Reference condition	N	GT	PURE	Gap
Primary: matched confidence=1 subset				
Original (speech-transcribed)	461	14.94	24.98	+10.04
Human back-translation (conf=1)	461	7.35	8.30	+0.95
Paired attenuation		9.08 BLEU pts (90.5%), CI $[7.44, 10.86]$		
Sensitivity: full 641 items				
Original (speech-transcribed)	641	12.78	23.02	+10.24 $[+8.89, +11.58]$
Human back-translation (full)	641	5.66	6.79	+1.12 $[+0.38, +1.91]$
Paired attenuation				9.89 BLEU pts (89.8%)
Human refs, reconstructions (6 seeds)	641	3.49–4.62	3.35–4.35	−0.15 to −0.65

Table 9: Asymmetric floor-effect control (full-641 sensitivity analysis): GT and PURE decode scores unde
human references. PURE consistently loses a larger absolute and fractional score than GT across all metri
simple symmetric compression (equal fractional score drop) an insufficient explanation. The matched-461 fra
ratio for BLEU-4 is 1.31× (GT 50.8% vs PURE 66.6%), consistent with the full-641 pattern shown here. Th
is descriptive; it does not discriminate between reference-text defects and training-distribution alignment.

Metric	GT (orig)	GT (human)	GT drop	PURE (orig)	PURE (human)	
BLEU-4	12.78	5.66	7.11 (55.7%)	23.02	6.79	17
BLEU-1	34.40	29.28	5.12 (14.9%)	55.43	32.35	23
chrF	34.59	32.75	1.84 (5.3%)	47.82	35.02	12
PURE/GT absolute-drop ratio	BLEU-4: 2.39× (17.00/7.11); BLEU-1: 4.51× (23.08/5.12); chrF: 6.97× (12.80/1					
PURE/GT fractional-drop ratio	BLEU-4: 1.28× (71.5%/55.7%); BLEU-1: 2.79× (41.6%/14.9%); chrF: 5.06× (26					
Dual-ref gap (orig+human)	BLEU-4: +11.50; BLEU-1: +20.03					

Absolute-drop ratio = (PURE drop in BLEU points) / (GT drop in BLEU points). Fractional-drop ratio = (PURE fractional drop %) /
drop %). The qualitative conclusion (PURE loses more than GT) is the same under either ratio; the paper's text uses the fractional ratio
"larger fraction" claims (1.28× for BLEU-4) and the absolute ratio when describing "2.4× more than GT" statements.

invert the advantage. Under the original SLRTP2025 evaluator, UNSEEN-PURE-v1 (retrieving from the 640 other test poses) scores 8.86 sacreBLEU versus 12.78 for GT, a gap of −4.27 BLEU points: the +10.24 advantage of SEEN-PURE-v1 does not survive retrieval-pool substitution. The donor-pool substitution estimand (SEEN − GT) − (UNSEEN − GT) = +14.51 BLEU points is large but is a *donor-pool substitution effect*, not a pure exposure effect, because SEEN-PURE-v1 and UNSEEN-PURE-v1 differ simultaneously in pool size (7,060 vs 640), median Jaccard (0.50 vs 0.37), donor reuse (Gini 0.11 vs 0.29), and training-pool exposure. Cross-fit controls

Table 10: Donor-origin contrast estimands (sacreBLEU, BLEU points; gaps vs local GT). SEEN = 7,060 training poses; UNSEEN = 640 other test poses. Per-seed cells in the artifact.

Evaluator	SEEN-PURE	UNSEEN-PURE	SEEN−GT	UNSEEN−GT	Estimand
Original	23.02	8.86	+10.24	−4.27	**+14.51**
Seeds (range)	6.83–9.09	6.59–7.98	−1.85 to −0.80	−1.94 to −0.27	−0.15 to +1.22

reproduce the same direction (CROSSFIT pooled −3.8 BLEU points vs fold-local GT). Under the six reconstructions all pool-origin contrasts lie in $[-0.15, +1.22]$ (Table 11).

Path-dependent decomposition and balanced factorial.

We decompose the substitution estimand descriptively along telescoping paths (SI Sup. E). The path-based origin estimate varies from +5.8 to +8.4 BLEU across admissible orderings because size and selection interact. The better-balanced 605-query factorial (matching seen/unseen donors at high/low LaBSE similarity within predeclared tolerances) estimates the pool-origin main effect at +2.08 $[+1.56, +2.57]$, the similarity main effect at +6.96 $[+5.93, +8.08]$, and the seen×similarity interaction at +3.70 $[+2.63, +4.79]$ (all Holm p=0.0003). Across reconstructions, the seen main effect shrinks to +0.51 and the interaction is null (−0.39). Two conclusions: Jaccard-only paths inflate the pool-origin component, and the released evaluator's seen advantage *grows with donor–query similarity* while reconstruction interactions are null. The caveat: seen donors remain more similar than unseen within strata (SMD +0.4), so this is a train-pool vs test-pool provenance contrast, not an identified training-exposure effect.

4.4 What distinguishes the released evaluator: five converging diagnostic views

We now turn the non-reproduction result into a positive characterization through five related diagnostic views (Fig. 3). These views are *not* independent confirmations: they reuse the same released checkpoint, PHX-public data, and related donor constructions, and are best read as triangulation on one signature. Table 12 summarizes the views (full version in SI Sup. D); the text gives the essential numbers.

Readout* with *generalization: the distinctive combination.

The released evaluator decodes its full 7,060-item training pool at free-decode BLEU 78.8 with 70.7% exact-match — strong readout — while retaining non-trivial dev/test performance (13.38/12.78). The family separates these regimes: validation-selected checkpoints show train ≈ dev BLEU (11.5 ≈ 9.9), while overfit rescue endpoints reach 99.9 train BLEU but collapse to dev NLL 4.90. Its donor-content pass-through is a 2.4× outlier (ratio 3.09 vs. family 1.05–1.31).

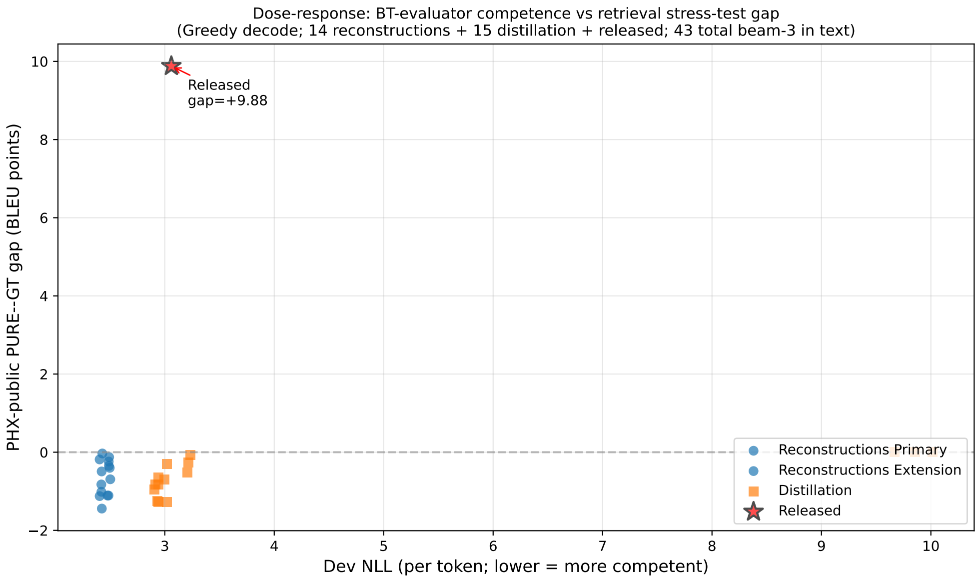

Fig. 3: PURE–GT gap versus dev competence (greedy decode; 14 reconstruction seeds + 15 distillation students + released evaluator = 30 plotted points). The x-axis is teacher-forced dev NLL (lower = more competent). An additional 13 checkpoints (ladder, config-faithful, step-faithful, confirmation, long-schedule, wd0 rescue) have beam-3 PHX-public gaps in the same range ($[-0.71, +0.00]$) but use a different dev-metric protocol and are not plotted; their values are in SI Sup. D. All 43 beam-3 evaluated checkpoints have gap ≤ 0. The figure is *exploratory*; the 43 points are NOT IID replicates. The released evaluator (star) sits far above the constructible family cloud at dev BLEU-4 13.38, gap +10.24. No synthetic or approximate points are used.

Leakage sanity checks for the train-pool readout.

Five checks (full data in SI Sup. N) rule out implementation-level artifacts: (i) the code-path audit confirms the decoder is autoregressive beam search with no teacher forcing; (ii) label permutation collapses BLEU from 78.6 to ∼1.0, confirming output–reference alignment; (iii) input-side pose permutation with output-equivariance verification ($E(\text{pose}_j, \text{slot}_i) = E(\text{pose}_j, \text{slot}_j)$ for 100/100 items, plus zero/noise/repeated-pose controls all collapsing BLEU to <1) rules out a pose-ignorant pure ID/cache lookup; (iv) exact-match rate is high across all 9 signers (59.0–74.8%) and even unique training texts (70.2% EM); and (v) a same-signer held-out control gives BLEU 12.19 / EM 3.2% on unseen test poses vs. 78.8 / 70.7% on training poses. These checks do not identify the training mechanism but are consistent with pose-conditioned training-pool exposure. The thresholds defining the signature were set after observing the released evaluator and are operational heuristics, not validated criteria (§3.3).

Table 11: Summary of the five diagnostic views. The released evaluator sits far outside the family range on all five views. Per-view family composition in SI Sup. D.

Diagnostic view	Family n	Released evaluator	Family range	Gap to family max
(i) Gap–competence	43	+10.24 at dev 13.38	$[-2.01, 0.00]$ at dev 4.5–10.9	+10.76
(ii) Pass-through ratio	24	3.09 (sentence)	1.05–1.31	+1.78
(iii) Train-pool readout BLEU	3	78.8 (70.7% EM)	9.2–11.5 (∼2% EM)	+67.3
(iv) Frame-reversal loss	1	−16.1 (PURE)	−4.3 (wd0 rescue)	|11.8|
(v) Competence gate	52	13.38	4.5–10.02 (max)	+3.36

Knowledge-distillation competence ladder.

To probe the unobserved competence interval above the reconstruction ceiling (dev BLEU-4 10.0), we trained 15 distillation students from the released evaluator as teacher: five distillation strengths ($\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$) crossed with three seeds, with loss $\mathcal{L} = (1-\alpha)\,\mathrm{CE} + \alpha\, T^2\, \mathrm{KL}(\text{teacher}\|\text{student})$ at T=2. All 15 completed training and beam-3 PHX-public evaluation. Table 13 shows the distilled students extend the observable range upward (dev BLEU-4 up to 10.9), yet all twelve α<1.0 students have negative PURE–GT gaps (−2.01 to −0.77). The three α=1.0 students produce empty or degenerate outputs (seed 101: 641/641 empty; seed 202: 483/641 empty, 158 single-token "."; seed 303: 641/641 empty), yielding near-zero GT and PURE BLEU and an approximately zero gap. Training-pool readout stays at 9.1–9.9 BLEU / 1.9–2.6% EM versus the released evaluator's readout. Full soft-label distillation (α=1.0) does not transfer the signature either. The readout-with-generalization signature is therefore not a property of the teacher's decoded output distribution that transfers through distillation.

Donor-registry consistency note.

The retrieval algorithm in §3.2 excludes donors whose NFKC-normalized text equals the query's. The headline PURE 23.79 (gap +11.01) predates this exclusion; applying it lowers PURE to 23.02 (gap +10.24) and was used for all 43 trained checkpoints. On 610/641 items (95.2%) both materializations produce identical output; for 30 of the 31 differing items, at least one excluded Jaccard=1.0 candidate whose decode matched the earlier hypothesis was found; one item remained unresolved (`results/donor_identification_31.json`). A 14-reconstruction sensitivity comparison shows mean $|\Delta\text{gap}| = 0.65$ BLEU (max 1.45).

Competence extrapolation (exploratory).

Three fits (OLS linear, OLS quadratic, GP with RBF + white kernel) over the 43 checkpoints spanning dev BLEU-4 $[4.5, 10.9]$ all predict negative gaps at 13.38, with substantial model sensitivity (OLS quadratic flips from −13.5 to +1.3 when distillation

Table 12: Per-α distillation ladder summary (15/15 beam-3 evaluated with canonical donor registry from §3.2, 15/15 with dev BLEU-4). α=0 = hard-label only; α=1 = full soft-label KL. All α<1.0 yield negative gaps; α=1.0 produces empty/degenerate outputs (641/641 or 483/641 empty hypotheses). See §4.4 for registry materialization consistency. Student train readout on first 1,000 training items for 12 students and on full 7,060 for three α=1.0 students; released evaluator readout on full 7,060 for direct comparison.

α	n	dev BLEU-4 (max)	gap (range)	train readout (BLEU / EM%)	reaches 10.5?
0.0 (hard-label)	3	10.51	$[-1.71, -0.80]$	9.1–9.8 / 2.1–2.6%	1/3
0.25	3	10.75	$[-2.01, -0.77]$	9.4–9.5 / 2.0–2.1%	2/3
0.5	3	10.41	$[-1.38, -0.77]$	9.2–9.9 / 2.1%	0/3
0.75	3	10.93	$[-1.21, -0.81]$	9.7 / 1.9–2.2%	2/3
1.0 (full KL)	3	10.81	$[0.00, 0.00]$	9.2–9.7 / 1.9%	1/3
Released (full 7,060, beam=3)	1	13.38	+10.24	78.82 / 70.7%	—

is removed; GP upper bound stays below +2.0 across leave-one-family-out folds). We treat this as an exploratory visual guide (full fit table in SI Sup. L), not as evidence that competence alone cannot explain the reversal. The unobserved 10.9–13.4 interval could in principle harbor non-linear behavior.

Synthesis.

Across the five diagnostic views, the released evaluator is the only checkpoint combining high training-pool readout (78.8 BLEU, 70.7% EM) with retained in-domain performance (13.38/12.78 dev/test). No reconstruction or distilled student reproduces either the +10.24 gap or this readout-with-generalization signature.

4.5 Checkpoint-targeted oracle stress tests (SI-diagnosed)

We treat evaluator-aware donor selection as a checkpoint-targeted oracle stress test, not as a deployable SLP method, and we report its unmasked advantage as a non-robust diagnostic: the diagonal advantage does not survive reference-overlap exclusion (SI Sup. H). The full cross-checkpoint transfer matrix, candidate-pool sweep, and selector/evaluator-label permutation test are likewise in SI Sup. H because they share target references and evaluator by design. The main text states only that the unmasked advantage is not robust to reference-overlap exclusion and that the white-box search budget (candidate-pool size) controls whether any decoded-BLEU margin is detectable.

5 Discussion

The audit yields a broad negative result (no constructible reconstruction or distilled student reproduces a gap remotely comparable to +10.24 across 43 beam-3 gap estimates (all 43 decoded with the canonical donor registry from §3.2 (95.2% decoded-output agreement with the earlier frozen materialization on the same queries; §4.4) (dev BLEU-4 4.5–10.9; all gap estimates are ≤ 0)) and a descriptive positive characterization: the released evaluator carries a high training-pool readout combined with retained in-domain performance that no public artifact in our family exhibits (§4.4). The 14 same-recipe reconstruction seeds (shared architecture, tokenizer, manifest, selection rule; seeds 101–1405) yield a Student-t prediction interval of $[-1.94, -0.46]$ BLEU points for the PURE–GT gap, while the heterogeneous families (distillation, ladder, config-faithful, step-faithful, rescue, confirmation) are reported separately as exploratory dose-response data, not as IID replicates of a single training process (SI Sup. D). Because all constructible checkpoints have lower competence than the released evaluator (family maximum dev BLEU-4 10.9 vs. released 13.38), this non-reproduction is observed under a competence confound — we cannot rule out that a nonlinear phenomenon emerges in the unobserved 10.9–13.4 interval; an exploratory OLS/GP extrapolation is reported in SI Sup. L for visualization only, without inferential claims. Because Saunders et al. Saunders et al. (2021) recommended BT for comparing between similar model configurations, and our evaluator family is additionally competence-unmatched (all checkpoints dev BLEU-4 < 10.9 vs. released 13.38), our non-reproduction result is scoped to the public training recipe and does not constitute a strong test of their rank-stability proposition.

On the matched 461-item confidence=1 subset, human-reference substitution descriptively attenuates the gap by 90.5% ($+10.03 \rightarrow +0.95$, paired bootstrap CI for attenuation $[7.44, 10.86]$); the asymmetric PURE-vs-GT drop ratio (1.29$\times$ fractional for BLEU-4) is inconsistent with a simple symmetric compression model but does not discriminate between reference-text defects and training-distribution alignment as non-exclusive candidate mechanisms. Czehmann et al. Czehmann et al. (2026) already showed that reference replacement changes BLEU scores and rankings on PHOENIX-2014T; our contribution on this axis is applying the matched-subset protocol to the released SLRTP2025 evaluator and quantifying paired attenuation, not establishing reference-set sensitivity as a novel phenomenon. The headline gap is robust to cluster-aware resampling: signer (9 clusters), broadcast-show (2 clusters: *heute*/*tagesschau*), broadcast-date (297 clusters), and signer$\times$show (17 clusters) bootstrap CIs are all consistent with the query-level CI, though the small cluster counts (especially show with only 2 clusters) make these exploratory sensitivity checks rather than confirmatory tests (SI Sup. G).

Relation to prior work.

These findings extend prior critiques of BT validity from output faithfulness to ranking stability: improved BT scores need not imply target conditioning Hong and Košecká, Jana (2026), low recorded-pose BT baselines are documented Walsh et al. (2024), and BT likelihood is a more stable correlate of human judgment than decoded BLEU Jiang et al. (2026). Our reversal is a *decoded-sacreBLEU* reversal; teacher-forced BT likelihood does not reproduce it (GT NLL 3.060 vs PURE 3.479), consistent with that divergence. Whole-donor replay is a constructed probe, whereas SOKE-style word-level retrieval can improve pose quality under a different representation Zuo et al. (2025); and multiple text scorers on the same BT decoding are not independent confirmation.

Validation frequency is an underappreciated hyperparameter.

Step-faithful vs epoch-validation shows validation frequency alone shifts the final checkpoint by 1.5–2.0 BLEU points (SI Sup. F).

Independent reproduction with greedy decoding.

Under greedy decode the released evaluator scores GT 12.24 and PURE 22.12 (gap +9.88); no trained checkpoint reproduces the reversal. Qualitative conclusions are identical to beam-3 (artifact at `results/cells_greedy/`).

What would be needed to identify the mechanism.

Identifying the mechanism requires the SLRTP2025 organizers to disclose (i) the exact training command and seeds, (ii) the data pipeline beyond [::2] subsampling, (iii) any curriculum or checkpoint averaging, and (iv) intermediate checkpoints. A leave-one-donor-out retrain sweep is computationally tractable ($\sim$1,400 GPU-hours) but requires the original training infrastructure, which is not publicly available.

Actionable recommendations for benchmark organizers.

These recommendations are scoped to the constructed replay-probe stress case on PHX-public, not to general SLP leaderboard stability.

- Release the training command, seeds, data pipeline scripts, and a SHA-256 manifest of training tensors (beyond the released `config.yaml`).
- Release intermediate checkpoints (at least every 100 epochs) so trajectory-level analyses can localize when distinctive properties emerge.
- Leaderboard entries that retrieve from the training pool should attach $\tau \in \{0.3, 0.5\}$ donor-exclusion results as a shortcut-robustness certificate.
- Adopt Czehmann et al. Czehmann et al. (2026) human back-translations as a secondary reference set and follow their matched-subset protocol (confidence=1 items) when comparing original-reference and human-reference conditions; flag systems whose gap diverges sharply between the two.
- Report training-pool free-decode BLEU ÷ dev free-decode BLEU (released: 6.1; reconstructions: 1.1–1.4) as a descriptive metric. Run the input-side pose permutation with fixed IDs and output-equivariance verification (SI Sup. N) to rule out a pose-ignorant pure ID/cache lookup.

6 Limitations

The audit is confined to PHX-public, a template-dense weather domain with documented reference-validity and train–test-overlap issues Czehmann et al. (2026); Alkain et al. (2026). The fixed nine-system stress-test set comprises replay, composition, weak generation, and random controls rather than competitive SLP systems, so the audit demonstrates a stress case on a released evaluator, not a general leaderboard-rank instability.

All 67 non-holdout checkpoints are less competent than the released evaluator (family maximum dev BLEU-4 10.9 vs. released 13.38); the original training history is unpublished, so competence-matched replication was not possible. The unobserved 10.9–13.4 interval could in principle harbor non-linear behavior. Saunders et al. Saunders et al. (2021) limited their rank-stability claim to similar model configurations; our result does not directly test their claim under those conditions. The specific mechanism is not identifiable from public artifacts: neither recipe-conditioned reconstruction, rescue sweeps, config-faithful retraining, nor knowledge distillation (any α, including full soft-label KL) transfers the property. The donor-pool substitution decomposition is path-dependent; pool-origin contrasts are descriptive associations, not identified training-exposure effects. The human back-translations are a single-annotator derivative resource whose high-confidence subset lacks second-annotator adjudication.

The reversal is a decoded-BT-text-metric reversal, not a sign-language quality reversal. Without target-conditioned DGS signer judgment or a validated sign-native pose metric, we cannot determine whether the retrieved donor motion conveys target-equivalent meaning; we explicitly do not claim that donor motion fails to express the target, nor that the reversal indicates dataset contamination or pose memorization. The DGS human evaluation is reported as an aborted feasibility study (Ethical Considerations); it does not enter the main empirical claims.

Use of public data.

All experiments use publicly released datasets (PHOENIX-2014T, CSL-Daily, SLRTP2025) and Czehmann et al.'s human back-translations. Pose tensors may retain signer identity; we release no per-signer raw poses.

7 Conclusion

This audit documents a +10.24 BLEU-point retrieval–ground-truth reversal on PHX-public that does not reproduce across 43 independently trained checkpoints (all decoded with a single canonical donor registry; range $[-2.01, 0.00]$). On the matched 461-item confidence=1 subset, human-reference substitution attenuates the gap by 90.5%. The released evaluator also decodes its training pool at 78.8 BLEU (70.7% EM) while retaining non-trivial dev/test performance, a signature no constructible checkpoint exhibits; input-side permutation and output-equivariance verification rule out a pose-ignorant ID/cache lookup. Its training log shows validation PPL rising from 14.84 at step 546 to 25.42 at the best-BLEU step 1820, indicating decoded-BLEU selection chose a checkpoint in a higher-NLL weight-space region that NLL-selected reconstructions never enter. The audit is scoped to the public training recipe: competence-matched replication was not possible, and the specific mechanism is not identifiable from public artifacts.

8 Compliance with Ethical Standards

Conflict of interest.

The authors declare no conflict of interest.

Ethical approval.

All experiments use publicly released datasets (PHOENIX-2014T, CSL-Daily, SLRTP2025) and Czehmann et al.'s human back-translations. No animal research was conducted.

Human participants.

A blind DGS human evaluation with 30 DGS-fluent raters was conducted as a feasibility study under a human-subjects exemption reviewed by the authors' institutional ethics committee (`[INSTITUTION-BLINDED] / [IRB-PROTOCOL-ID-BLINDED]`). Raters provided informed consent; only aggregate statistics are reported. The video files and UUID-to-system manifest were lost during a host-cleanup operation (2026-07); per-rater response CSVs are preserved in the artifact. The study is reported as an aborted feasibility study, not as audit evidence. No Deaf community member was involved in the audit design.

Funding.

Anonymous.

Data availability.

All numerical values, per-checkpoint data, decoded hypotheses, and analysis scripts are in the anonymous artifact at https://anonymous.4open.science/r/rcp-audit-artifact-B314/README.md.

Supplementary Information (SI)

The following supplementary materials, referenced throughout the main text, are available as a separate file (`supplementary.pdf`) and in the anonymous artifact:

- Sup. A: Missing Test ID Sensitivity (`sec:appendix_missing_id`).
- Sup. B: Per-seed Extension Cells 707–1405.
- Sup. C: Analysis Provenance and Multiplicity (`sec:provenance`).
- Sup. D: Canonical Checkpoint Registry details (`tab:checkpoint_registry`).
- Sup. E: Donor-Pool Substitution Decomposition (`sec:appendix_decomposition`, `tab:path_sensitivity`).
- Sup. F: Config-Faithful and Perturbation Details (`sec:appendix_config_faithful`, `sec:appendix_perturbation`, `tab:perturbation`).
- Sup. G: Cluster-Aware Bootstrap (`sec:appendix_cluster_bootstrap`).
- Sup. H: Supplementary Oracle Diagnostics (`sec:appendix_oracle`).
- Sup. I: Template-Density, Holdout Boundary, Donor-Exclusion (`sec:appendix_template_boundary`).
- Sup. J: DGS Human Evaluation Details (`sec:appendix_human_eval`).
- Sup. K: Transcript-Slot Diagnostic Details (`sec:appendix_slot`).
- Sup. L: Competence Extrapolation Details (`sec:appendix_extrapolation`).
- Sup. M: CSL-Daily Boundary Control (`sec:csl-daily`).
- Sup. N: Leakage Sanity Checks for Train-Pool Readout (`sec:appendix_leakage`).

Checkpoint Registry (summary)

Table 13: Canonical checkpoint registry. 70 planned training runs, all 70 completed. 67 non-holdout runs have dev metrics; 43 have both dev BLEU-4 and PHX-public PURE–GT gap under the canonical §3.2 registry. Full per-checkpoint details in SI Sup. D.

Family	Planned	Trained	Has gap	Has dev	Enters
Reconstructions (14)	14	14	14	14	multiseed, bootstrap, PI, dose–resp.
Rescue (20)	20	20	1	20	gate; wd0 dose–resp.
Other families (21)	21	21	14	20	dose–resp.
Distillation (15)	15	15	15	15	ladder, dose–resp.
Total	70	70	43	67	
Released original	—	—	1	1	audit target

References

Alkain, B., A. Núñez-Marcos, C. Escolano, L. Docío-Fernández, O. Perez-de Viñaspre, and G. Labaka. 2026. Critical analysis of datasets for sign language translation. *Frontiers in Artificial Intelligence* 9: 1743223. https://doi.org/10.3389/frai.2026.1743223 .

Artiaga, K., S. Kamila, H. Afli, C. Lynch, and M. Hasanuzzaman 2025. Rethinking sign language translation: The impact of signer dependence on model evaluation. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, Suzhou, China, pp. 18379–18391. Association for Computational Linguistics.

Camgoz, N.C., S. Hadfield, O. Koller, H. Ney, and R. Bowden 2018. Neural sign language translation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 7784–7793.

Carlini, N., F. Tramer, E. Wallace, M. Jagielski, A. Herbert-Voss, K. Lee, A. Roberts, T. Brown, D. Song, U. Erlingsson, A. Oprea, and C. Raffel 2021. Extracting training data from large language models. In *30th USENIX Security Symposium (USENIX Security 21)*, pp. 2633–2650.

Cory, O., M. Ivashechkin, K. Sahin, O. Ranum, J. Low, E. Fish, A. Pelykh, et al. 2026. BackTranslation2.0: A linguistically motivated metric to assess sign language translation. *arXiv preprint arXiv:2606.28673*. arXiv:2606.28673 .

Czehmann, V., S. Yazdani, Y. Hamidullah, F. Nunnari, and E. Avramidis 2026. "A Sacred Bird Called the Phoenix": Auditing the most-used parallel corpus for German sign language recognition and translation. In *Proceedings of the LREC2026*

12th Workshop on the Representation and Processing of Sign Languages: Language in Motion, Palma, Mallorca, Spain, pp. 80–92. ELRA Language Resources Association. ISBN 978-2-493814-82-1. Back-translated test set and annotations: https://github.com/DFKI-SignLanguage/sacre-bird-phoenix (CC BY-NC-SA 4.0).

Desai, A., M. De Meulder, J.A. Hochgesang, A. Kocab, and A.X. Lu 2024. Systemic biases in sign language AI research: A Deaf-led call to reevaluate research agendas. In *Proceedings of the LREC-COLING 2024 11th Workshop on the Representation and Processing of Sign Languages: Evaluation of Sign Language Resources*, Torino, Italy, pp. 54–65. ELRA and ICCL.

He, T., K. Rahul, D. Hupkes, and R. Cotterell 2024. On the blind spots of model-based evaluation metrics for text generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics*.

Hong, R. and Košecká, Jana. 2026. Conditional collapse in sign language production: A diagnostic and a scaling argument. *arXiv preprint arXiv:2606.01643*. arXiv:2606.01643 [cs.CV].

Imai, S. 2025. SiLVERScore: Why aren't semantically-aware embeddings used for sign language generation evaluation? In *Proceedings of the International Conference on Recent Advances in Natural Language Processing (RANLP 2025)*, pp. 519–528.

Jiang, Z. et al. 2026. Meaningful pose-based sign language evaluation. In *Proceedings of the 10th Workshop on Machine Translation (WMT)*.

Kandpal, N., E. Wallace, and C. Raffel 2022. Looking inside my memory: Quantifying memorization in language models. In *Findings of EMNLP*.

Khandelwal, U., A. Fan, D. Jurafsky, L. Zettlemoyer, and M. Lewis 2020. Generalization through memorization: Nearest neighbor language models. In *International Conference on Learning Representations (ICLR)*.

Kim, J.H., M. Huerta-Enochian, C. Ko, and D.H. Lee 2024, May. SignBLEU: Automatic evaluation of multi-channel sign language translation. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, Torino, Italia, pp. 14796–14811. ELRA and ICCL.

Post, M. 2018. A call for clarity in reporting BLEU scores. In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pp. 186–191.

Saunders, B., N.C. Camgoz, and R. Bowden 2020. Progressive transformers for end-to-end sign language production. In *European Conference on Computer Vision*.

Saunders, B., N.C. Camgöz, and R. Bowden. 2021. Continuous 3D multi-channel sign language production via progressive transformers and mixture density networks. *International Journal of Computer Vision* 129: 2113–2135. https://doi.org/10.1007/s11263-021-01457-9 .

Sellam, T., D. Das, and A. Parikh 2020. BLEURT: Learning robust metrics for text generation. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 7881–7892.

Tasyurek, S.M., T. Kiziltepe, and H.Y. Keles. 2025. Disentangle and regularize: Sign language production with articulator-based disentanglement and channel-aware regularization. *arXiv preprint arXiv:2504.06610*. arXiv:2504.06610 .

Testa, Z., A. Furnari, L. Baraldi, and N. Díaz-Rodríguez 2026. SLU-2K: A question-based benchmark for semantic evaluation of sign language translation. In *GenSign Workshop, CVPR.* arXiv:2606.03788.

Walsh, H., E. Fish, O. Mercanoglu Sincan, M.I. Lakhal, R. Bowden, N. Fox, B. Woll, K. Wu, Z. Li, W. Zhao, H. Wang, W. Zhou, H. Li, S. Tang, J. He, X. Wang, R. Zhang, Y. Wang, L. Cheng, M. Tasyurek, T. Kiziltepe, and H.Y. Keles 2025. SLRTP2025 sign language production challenge: Methodology, results, and future work. In *Proceedings of the Third Sign Language Recognition, Translation and Production Workshop at CVPR 2025.* arXiv:2508.06951.

Walsh, H., A. Ravanshad, M. Rahmani, and R. Bowden 2024. A data-driven representation for sign language production. In *British Machine Vision Conference.*

Walsh, H., B. Saunders, and R. Bowden 2024. Sign stitching: A novel approach to sign language production. In *British Machine Vision Conference.*

Yazdani, M. et al. 2026. A critical study of automatic evaluation in sign language translation. In *Proceedings of LREC-COLING 2026.*

Zhang, T., V. Kishore, F. Wu, K.Q. Weinberger, and Y. Artzi 2020. BERTScore: Evaluating text generation with BERT. In *International Conference on Learning Representations (ICLR).*

Zuo, R., R.A. Potamias, E. Ververas, J. Deng, and S. Zafeiriou 2025. Signs as tokens: A retrieval-enhanced multilingual sign language generator. In *Proceedings of the IEEE/CVF International Conference on Computer Vision.*